

Khush A. Patel

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Objective

Mechanical Engineering student at Johns Hopkins University with research and industry experience in computational engineering, and electromechanical development.

Education

Johns Hopkins University, Whiting School of Engineering

Baltimore, MD

B.S. Mechanical Engineering **GPA: 4.00/4.00**

Anticipated May 2028

Relevant Coursework: Statics, Mechanics, Thermodynamics, Differential Equations, Linear Algebra, Multivariable Calculus, Electricity & Magnetism, Digital Electronics, CAD (SolidWorks, Creo), Python.

Technical Skills

- **CAD & Engineering Analysis:** SolidWorks, PTC Creo, Windchill, ANSYS Mechanical, Fluent, CFX, OpenFOAM, Linux
- **Programming:** Python (NumPy, SciPy), MATLAB, C++, MathCAD, Git, Excel
- **Robotics & Controls:** ROS 2, Kinematics, Motion Planning, Trajectory Control, PID, LQR, Model Predictive Control (MPC)
- **Mechanical Design:** GD&T, RSS & Worst-Case Tolerance Stack-Up Analysis, DFM/DFA, DFMEA, Technical Drawings
- **Manufacturing:** CNC Milling, CNC Turning, Wire EDM, Sheet Metal Fabrication, FDM & Resin 3D Printing, Laser Cutting

Engineering Experience

Amphenol

Valley Green, PA

Development Engineering Intern – High Speed I/O Connectors (Current)

May 2026–Aug 2026

- Designed **10+** high-speed connector components in **SolidWorks** for next-generation AI and hyperscale data-center applications.
- Simplified non-critical geometries across **5+** sheet metal and CNC-machined components, reducing material usage by up to **15%** while applying **DFM** principles to improve manufacturability and assembly speed; validated all design changes against failure modes using **ANSYS Static Structural**.
- Conducted **GD&T** and statistical tolerance stack-up analyses on **5+ production drawings**, selecting **ISO hole/shaft fit classes** and evaluating **MMC/LMC** conditions using **RSS (6 sigma)** and **worst-case** methodologies to ensure manufacturability and assembly.

Johns Hopkins University – AIRO Laboratory

Baltimore, MD + Remote

Research Assistant – Computational Aerodynamics & Controls (Current)

Dec 2025–Present

- Ran **20+** high-fidelity **OpenFOAM LES high Reynold simulations (2M+ cell meshes)** to model the air wake of naval vessels.
- Built Python pipelines to compress flow-field data into control-oriented reduced-order models (ROMs), matching full CFD results to **95%** accuracy while cutting data footprint by **600x (120 GB → 200 MB)**.
- Delivered these ROMs directly into the lab's **Model Predictive Control (MPC)** framework for autonomous UAV landing research, bridging high-fidelity aerodynamics with real-time, control-ready models.

Amarkosha Technologies (Start-Up)

Baltimore, MD

Mechanical Lead – Product Development

Jul 2025–Dec 2025

- Led mechanical development of a vortex-induced vibration energy harvester for low-speed wind environments, iterating through **20+** design cycles on the coil-magnet architecture to improve conversion efficiency **3x**, reaching **30W** output at **70%** generator conversion efficiency.
- Validated structural margins via transient CFD, fluid-structure interaction (FSI), and modal analyses in **ANSYS Fluent, CFX, and Mechanical**, confirming max deflection remained within threshold for **T300 carbon fiber** under target operating loads.
- Selected for the Johns Hopkins SPARK Accelerator; project named a BII Grant finalist.

Johns Hopkins Mars Rover Team

Baltimore, MD

Mechanical & Controls Engineer – Manipulator

Sep 2025–Dec 2025

- Derived an equivalent hand-calculated torque model across all **4** wrist motors to size actuators, enabling selection of **BLDC** motors at **50%** lower cost than baseline while meeting torque requirements; achieved **7kg** payload capacity, cross-validated via experimental testing and hand calculations.
- Designed housing under **DFM** principles, using **SolidWorks static FEA** to evaluate shaft-coupling stress across multiple thicknesses (**1.3** safety factor), reducing housing weight by approx **20%**.

Projects

4-DOF Robotic Arm and In Depth ANSYS Static Study

- Designed, built, and controlled a 4-DOF robotic arm featuring Arduino-based actuation and closed-loop encoder feedback, validated with **ANSYS FEA** and hand calculations, on a **\$30** budget; achieved 0.0025 max strain under **400g** payload capacity. Mechanical prototype built in **3 weeks**, full hardware/software control integration in **1** additional week (**4 weeks** total).